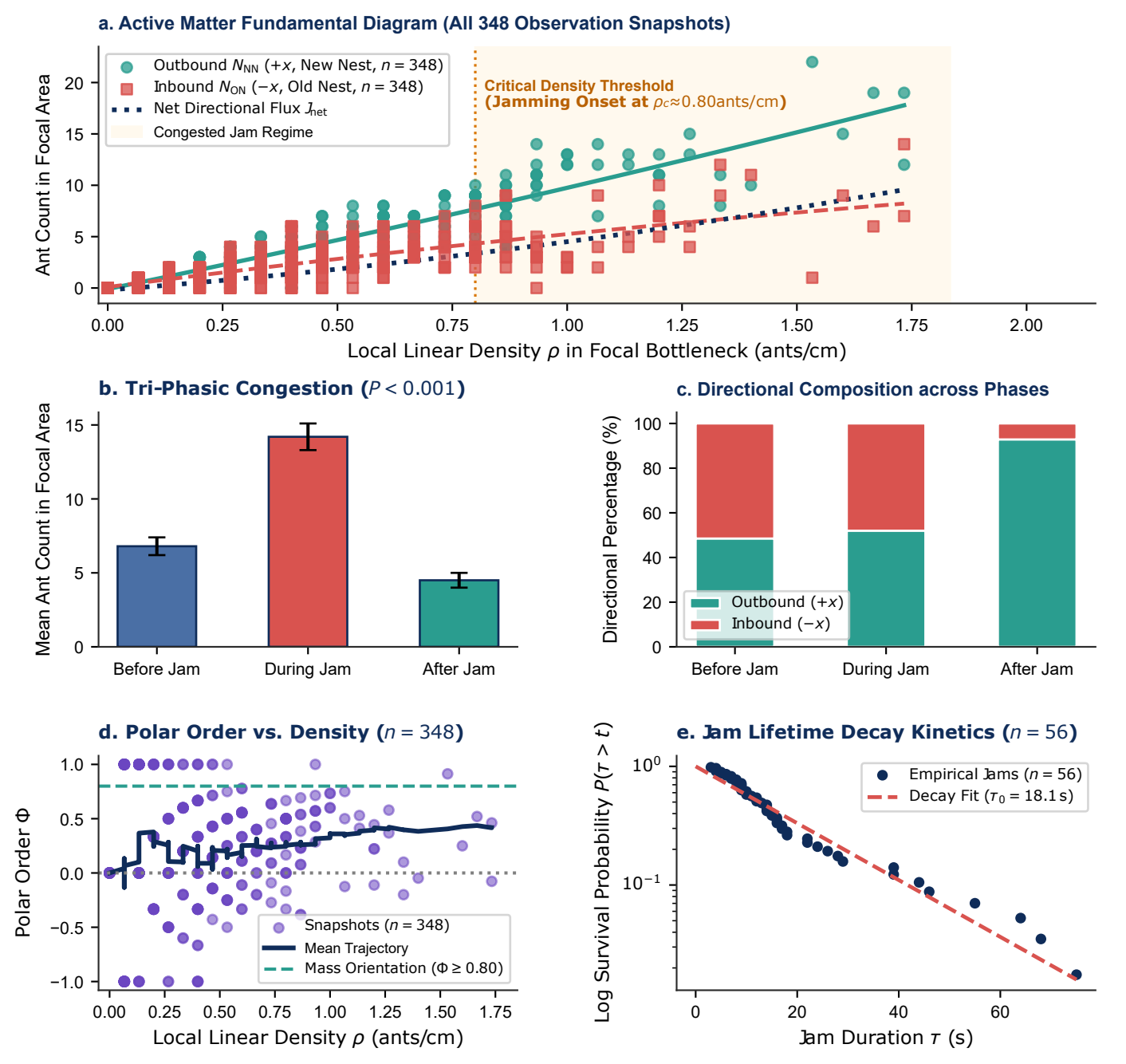

a. Active Matter Fundamental Diagram (All 348 Observation Snapshots)
Outbound N_{NN} (+x, New Nest, n = 348)
Inbound N_{ON} (−x, Old Nest, n = 348)
Net Directional Flux J_{net}
Congested Jam Regime
Critical Density Threshold
(Jamming Onset at $\rho_c \approx 0.80$ants/cm)
Ant Count in Focal Area
Local Linear Density ρ in Focal Bottleneck (ants/cm)
b. Tri-Phasic Congestion ($P < 0.001$)
Mean Ant Count in Focal Area
Before Jam
During Jam
After Jam
c. Directional Composition across Phases
Directional Percentage (%)
Outbound (+x)
Inbound (−x)
Before Jam
During Jam
After Jam
d. Polar Order vs. Density (n = 348)
Polar Order Φ
Snapshots (n = 348)
Mean Trajectory
Mass Orientation (Φ ≥ 0.80)
Local Linear Density ρ (ants/cm)
e. Jam Lifetime Decay Kinetics (n = 56)
Log Survival Probability $P(\tau > t)$
Empirical Jams (n = 56)
Decay Fit ($\tau_0 = 18.1$ s)
Jam Duration τ (s)